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# NRC7394 Evaluation Kit User Guide

## (Web GUI)

**Ultra-low power & Long-range Wi-Fi**

**Ver 1.0**
**Sep. 28, 2026**

## NEWRACOM, Inc.

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**NRC7394 Evaluation Kit User Guide (Web GUI)**
**Ultra-low power & Long-range Wi-Fi**


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# 1 Overview and prerequisites

## 1.1 Intended Use and Software Disclaimer

The Web GUI is provided for evaluation and demonstration with the NRC7394 Evaluation Kit (EVK). Its source code is provided as a reference implementation for development and is not validated for production deployment.

The Web GUI is an optional component of this package. On a board where it is not installed, configure and start the wireless function with start.py as before.

Developers who incorporate or adapt this software for their own products are responsible for product integration, testing, security hardening, compliance with applicable requirements, and ongoing maintenance, including security updates.

Except as expressly provided in a separate written agreement with Newracom, and to the extent permitted by applicable law, the Web GUI and its source code are provided "AS IS," without warranties of any kind, express or implied, including warranties of merchantability or fitness for a particular purpose. To the extent permitted by applicable law, Newracom shall not be liable for damages arising from their use, modification, or deployment.

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## 1.2 Using the Web GUI

The Web GUI lets you change the wireless settings of a board and check its link state from a browser. It covers the operating mode such as AP or STA, the network and the wireless features, without editing configuration files or using the command line. It manages the state and the settings of a single board.

To reach it you need a board on which the Web GUI is installed and running, and a PC that can reach that board. Use the wired management address printed when `install.sh` finishes. Section 2 describes the installation.

Use the **wired management connection** for configuration work. A HaLow link can drop while wireless settings are applied, and changing the wired management address itself also changes the address of the GUI.

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# 2 Installation

## 2.1 Package and prerequisites

The Web GUI is installed by `install.sh` in the `evk/sw_pkg` directory of the release package. The earlier `update.sh` copies only `nrc_pkg/` and does not install the Web GUI.

Before installing, check the following on the board.

- The board is a Raspberry Pi with the NRC7394 EVK connected and the standard HaLow package environment set up. The S1G-capable hostapd and wpa_supplicant used by the HaLow stack must already be installed; install.sh does not install them.
- The kernel headers for the running kernel and the build tools are installed, or the board has internet access so that `install.sh` can install them. See Section 2.3 for a board without internet access.
- The board is reached over its wired port. The HaLow link can drop while the package is installed.

**Table 2.1 Installation methods**

| Method | Command | Driver | Web GUI |
|---|---|---|---|
| Full package (recommended) | `sudo ./install.sh` | Built on the board from the src directory of the package | Installed |
| `evk/sw_pkg` only | `sudo ./install.sh --no-build` | Prebuilt nrc.ko in the package. The board kernel must match the kernel it was built for, and the device-tree overlay must already be in place from an earlier installation | Installed |
| Legacy update | `./update.sh` | Prebuilt | Not installed |

Use the full package unless the board kernel is known to match the prebuilt driver. With `--no-build`, a kernel mismatch is reported as a warning at install time and the radio fails to start later. The device-tree overlay is not installed with --no-build either, so use this option only on a board where a full installation has already been done. Check the kernel first:

```
modinfo -F vermagic evk/sw_pkg/nrc_pkg/sw/driver/nrc.ko
uname -r
```

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## 2.2 Installing

1. Copy the whole release package to the board, for example with `scp -r`, and open `evk/sw_pkg` in it.
2. Run `sudo ./install.sh`. Add `--no-deps` on a board without internet access (Section 2.3).
3. Reboot the board if the installer says a reboot is required.
4. Note the wired management address printed at the end of the installation. It is the address of the Web GUI (Section 3.1).

```
cd ~/nrc7394_sw_pkg_v<version>/evk/sw_pkg
sudo ./install.sh
```

What `install.sh` does:

- Installs the build and runtime dependencies from the distribution package repository.
- Copies `nrc_pkg/` to `~/nrc_pkg`. An existing `~/nrc_pkg` is moved to `~/nrc_pkg_backup_<date>`.
- Builds the driver and cli_app for the running kernel, and installs the device-tree overlay on a Raspberry Pi.
- Installs the systemd services of the HaLow stack (`halow.target`, `halow-wlan`, `halow-hostapd`, `halow-webd`) and enables them at boot.
- Starts the Web GUI daemon `halow-webd`.

**Installation does not start the wireless function.** Only the Web GUI daemon is started, so an installed board does not transmit until the wireless function is started from the dashboard or Start radio at boot is enabled under System in Settings.

Existing Web GUI settings are kept. The installer prints `kept existing /etc/halow/halow.conf` when it found a previous configuration and `created /etc/halow/halow.conf` when it started from the initial state. To install with factory settings, remove `/etc/halow` before running the installer. The administrator password then returns to `admin` and a new certificate is generated.

## 2.3 Installing on a board without internet access

Run the installer with `--no-deps` to skip the package repository step. The installer then relies on what is already on the board: the kernel headers for the running kernel, the C build tools, `cmake`, `libnl-genl-3-dev`, `libcjson-dev`, `libreadline-dev`, `python3`, `iw`, `net-tools`, `bridge-utils` and a DHCP client. A missing item stops the installation part-way or leaves a board on which the radio does not start.

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Either stage those packages while the board still has a network, or download them on a machine with the same distribution and copy the package files to the board.

```
sudo dpkg -i /path/to/debs/*.deb
cd ~/nrc7394_sw_pkg_v<version>/evk/sw_pkg
sudo ./install.sh --no-deps
```

If the prebuilt driver matches the board kernel, `--no-build` avoids the build tools altogether:

```
sudo ./install.sh --no-deps --no-build
```

After the installation, confirm that the Web GUI daemon is running and answers:

```
systemctl is-active halow-webd
curl -sk -o /dev/null -w '%{http_code}\n' https://127.0.0.1:8443/    # expect 200
journalctl -u halow-webd -n 50    # daemon log if it does not answer
```

## 2.4 Installed locations

**Table 2.2 Installed locations**

| Location | Contents |
|---|---|
| `~/nrc_pkg/` | Scripts, configuration files, firmware, and the prebuilt driver and cli_app |
| `/usr/lib/halow/` | Helper scripts and `uninstall.sh` |
| `/usr/lib/halow/web/` | The Web GUI daemon and its pages |
| `/etc/halow/` | Web GUI settings: `config.json` (including the administrator password), `halow.conf`, `schema.json` and the TLS certificate |
| `/etc/systemd/system/` | `halow.target` and the HaLow service units |

The Web GUI runs from `/usr/lib/halow/web/`. Changes made to `~/nrc_pkg/web/` are not used. After editing a file under `/usr/lib/halow/web/`, restart the daemon with `sudo systemctl restart halow-webd`.

## 2.5 Uninstalling

```
sudo /usr/lib/halow/uninstall.sh          # keeps settings and ~/nrc_pkg
sudo /usr/lib/halow/uninstall.sh --purge    # also removes settings
```

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Without options, the uninstaller removes the systemd services, `/usr/lib/halow` and the loaded kernel module, and keeps `/etc/halow` and `~/nrc_pkg` so that a later installation continues with the same settings. `--purge` also removes the settings, the administrator password and the certificate, `~/nrc_pkg`, the firmware files under `/lib/firmware` and the HaLow logs. Add `--dts` to remove the device-tree overlay as well; a reboot is then required. Backup directories `~/nrc_pkg_backup_*` are never removed.

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# 3 Quick start

## 3.1 Connecting and logging in for the first time

1. Open `https://<board address>:8443` in a browser. Replace `<board address>` with the wired management address of the board.
2. If a self-signed certificate warning appears, confirm that the address is the intended board and continue according to the browser's procedure.
3. Enter the initial administrator password admin and select Sign in.
4. Change the administrator password under **Settings → System**. It must be at least 5 characters; use a value other than the default.
5. Configure the wireless settings after the password change is complete.

No other setting can be saved before the password change is complete. If `change the admin password first` appears, complete the password change first.

Connecting on HTTP port 8080 redirects to HTTPS port 8443.

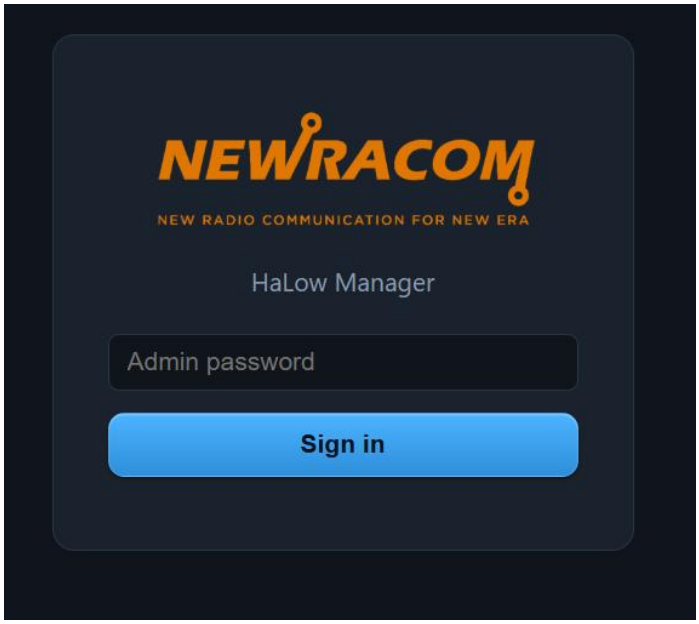



**Figure 3.1 Administrator sign-in page**

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## 3.2 Starting an AP

1. Select AP mode and the country for the installation site under **Settings → General**.
2. Set the SSID, the security method and the password under **Network**.
3. Select the channel width and the channel. Where automatic selection is supported, **Channel** can be set to `auto`.
4. Check the address settings and the bridge configuration under **LAN**.
5. Press **Apply** to save the settings and follow the application notice.
6. If the wireless function is stopped, start it from the dashboard.
7. Check the run state, the SSID and the actual channel on the dashboard, then connect a STA.

`auto` cannot be used in relay mode, on an AP that has a connected relay, or under country code K2. A relay stays on its upstream AP's channel, and the current software package does not provide a function that informs a relay of a channel change on the AP. In those cases set the channel explicitly as the screen indicates.

## 3.3 Connecting as a STA

1. Select STA mode and the country under **Settings → General**.
2. Enter the SSID, the security method and the password of the AP to join, under **Network**.
3. Check the address settings under **LAN**.
4. Save with **Apply**, and start the wireless function from the dashboard if it is stopped.
5. Check the connection state on the dashboard and verify actual data traffic.

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# 4 Changing and applying settings

## 4.1 Finding a setting

Only the items available in the current operating mode are shown. If an item is missing, check the operating mode first.

**Table 4.1 Settings areas**

| Area | Main settings |
|---|---|
| General | Operating mode, country |
| Network | SSID, security, password, channel, channel width, beacon interval, channel manager |
| LAN | Bridge, wired port, IP address, DHCP server range, static addresses |
| Radio | TX power, TPC, guard interval, AMPDU, CCA threshold, short beacon |
| Power | Power save, listen interval, BSS max idle, sleep duration, duty cycle |
| System | Administrator password, wireless autostart at boot, MAC address, CPU clock, log |

Set the operating mode, the country, and the network and address settings first. Change the remaining wireless and power items according to your purpose. For TPC and the channel manager in detail, see their own user guides.

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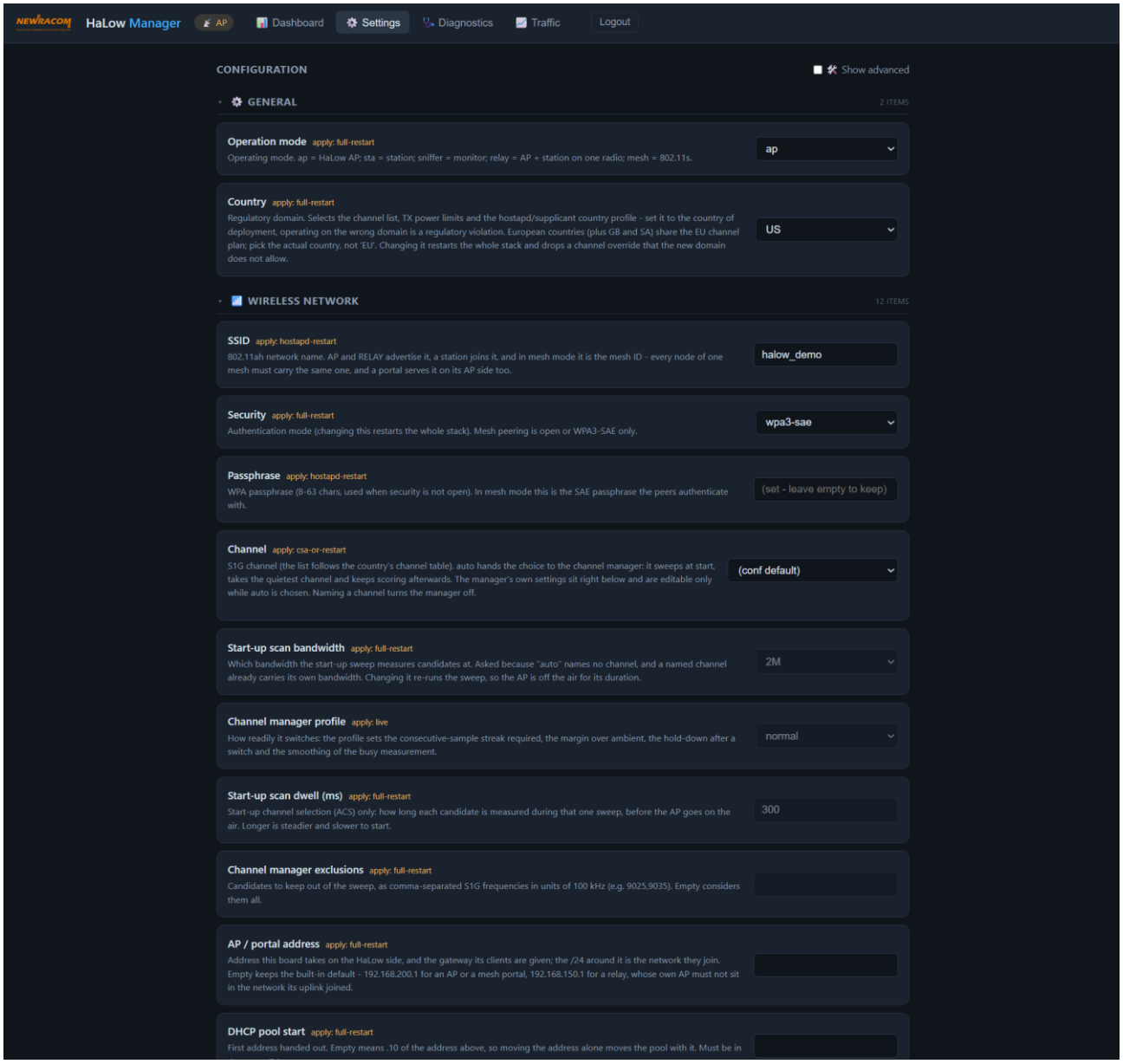



**Figure 4.1 Settings page showing General and Wireless Network**

## 4.2 Saving and checking the result

1. Change the items you need.
2. Press **Apply** and review how the changes will be applied.
3. If an error is shown, correct that item and apply again.
4. When the changes have been applied, check the actual state on the dashboard.

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If input validation finds an error, that save request is not applied. Combinations that cannot be used together, such as a fixed channel with automatic channel selection, are also reported on the screen with the reason.

Pressing **Apply** with nothing changed asks whether to restart with the saved settings.

## 4.3 When the connection drops while settings are being applied

Depending on the items changed, the settings take effect immediately or the wireless service is restarted. Use the table below to read the result shown on the screen.

**Table 4.2 How a change is applied**

| Shown | Effect on the user |
|---|---|
| `none` | Settings are saved only, nothing is restarted |
| `live` | Applied immediately to the running wireless function |
| `hostapd-restart` | Restarts the network service that matches the role, or reloads its configuration. A reconnection may be needed; in mesh this is a full restart |
| `csa-or-restart` | Switches the channel while running if possible. On failure it may fall back to a full restart without a further prompt |
| `full-restart` | Full restart of the wireless function. Typically about 1 to 3 minutes |
| `settings-only` | Saved while the wireless function is stopped. Applied at the next start |
| `eth-config` | Wired settings applied. The management connection can drop and the GUI address can change |
| `autostart` | Change to the wireless autostart-at-boot setting |

When several items are changed together, the most disruptive restart required by those changes is used. The wired settings and the autostart setting are applied independently of the wireless run state.

A full restart continues on the board even if the browser is closed. You can connect again and check the current state. If you were connected over the HaLow link, connect over the wired management address.

## 4.4 Using it together with the command line

Settings saved in the GUI are also written to `start.py` and the related configuration files, so they are used by a later manual run. Settings used when you run `start.py` manually are also reflected in the GUI. If the wireless function is running under the Web GUI, a manual start.py or stop.py run first stops the Web GUI's wireless service (halow.target). To return to the Web GUI, run sudo systemctl start halow.target or start the wireless function from the dashboard.

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If the two disagree, check the actual state on the dashboard first. Writing the configuration file may have failed, so collect the logs according to the troubleshooting procedure.

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# 5 Status and tools

## 5.1 What to check on the dashboard

**Table 5.1 Dashboard items**

| Purpose | Item |
|---|---|
| Whether the wireless function is running | Run state. It can also be started or stopped here |
| Whether this is the network you configured | Operating mode and SSID |
| The channel actually in use | Current channel and channel width |
| Transmit power | TX power and TPC state |
| Versions to give to technical support | Firmware and driver versions |

Do not assume the settings have taken effect based on the settings screen alone; check the actual state on the dashboard and verify traffic as well.

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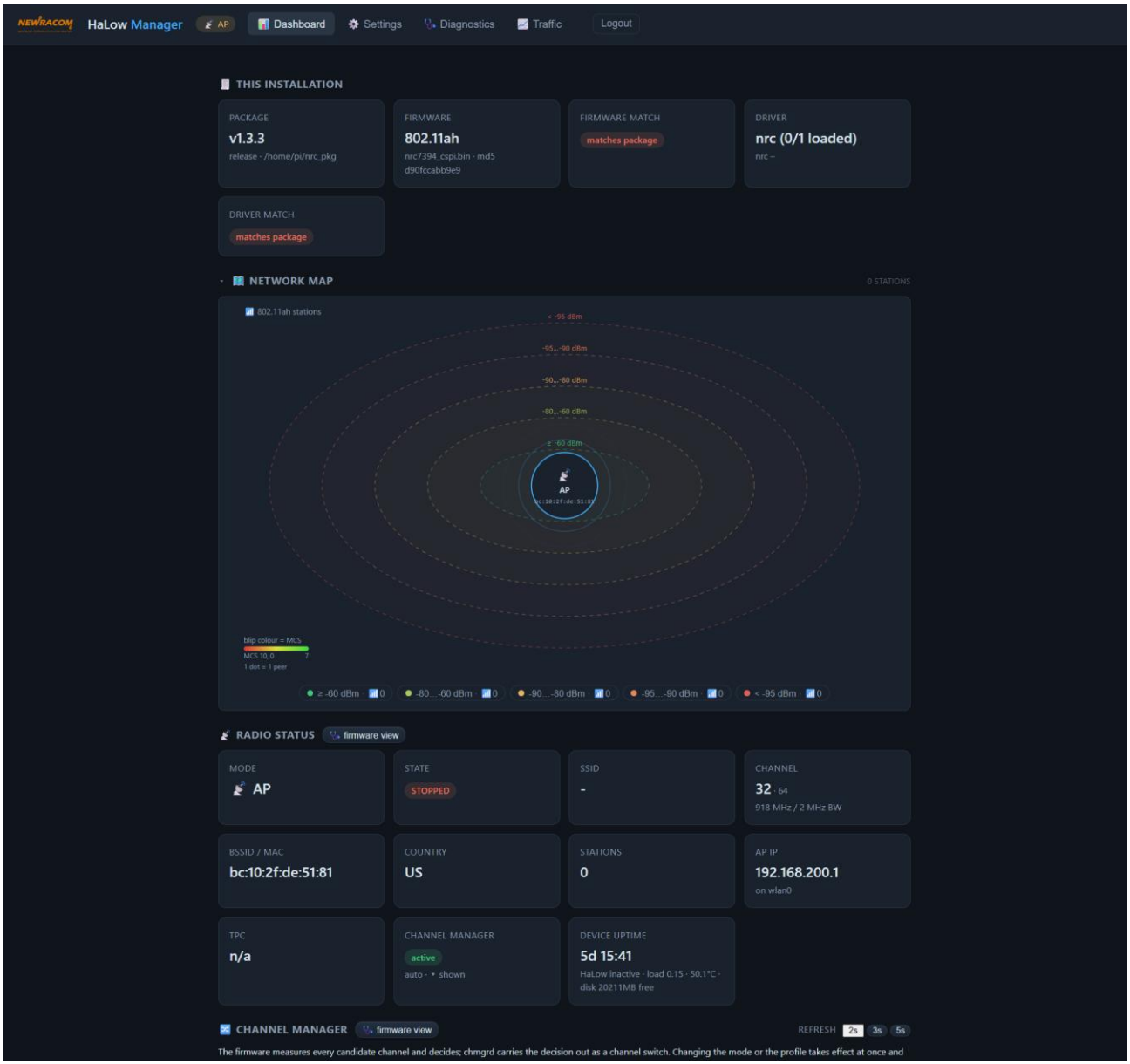



**Figure 5.1 Dashboard overview with installation, network map and radio status**

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## 5.2 Checking connected devices

In AP and relay modes, the STA table shows the MAC address, RSSI, MCS and connection time. Devices discovered behind a bridge are listed in a separate table.

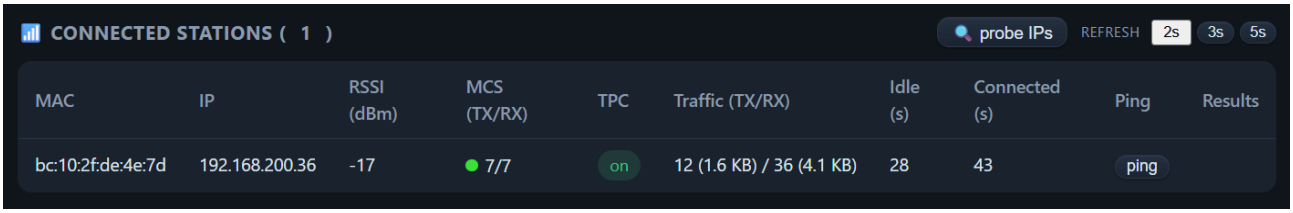



**Figure 5.2 Connected stations shown on an AP**

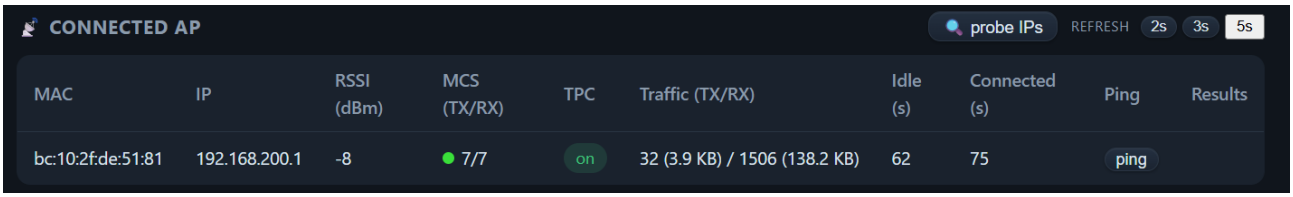



**Figure 5.3 Connected AP shown on a STA**

The network map places the STAs in RSSI bands. The position means a signal-strength band, not an actual distance. The color follows the modulation in use. Clicking the badge of a band filters the table to that band only. Devices behind a bridge are drawn as small squares.

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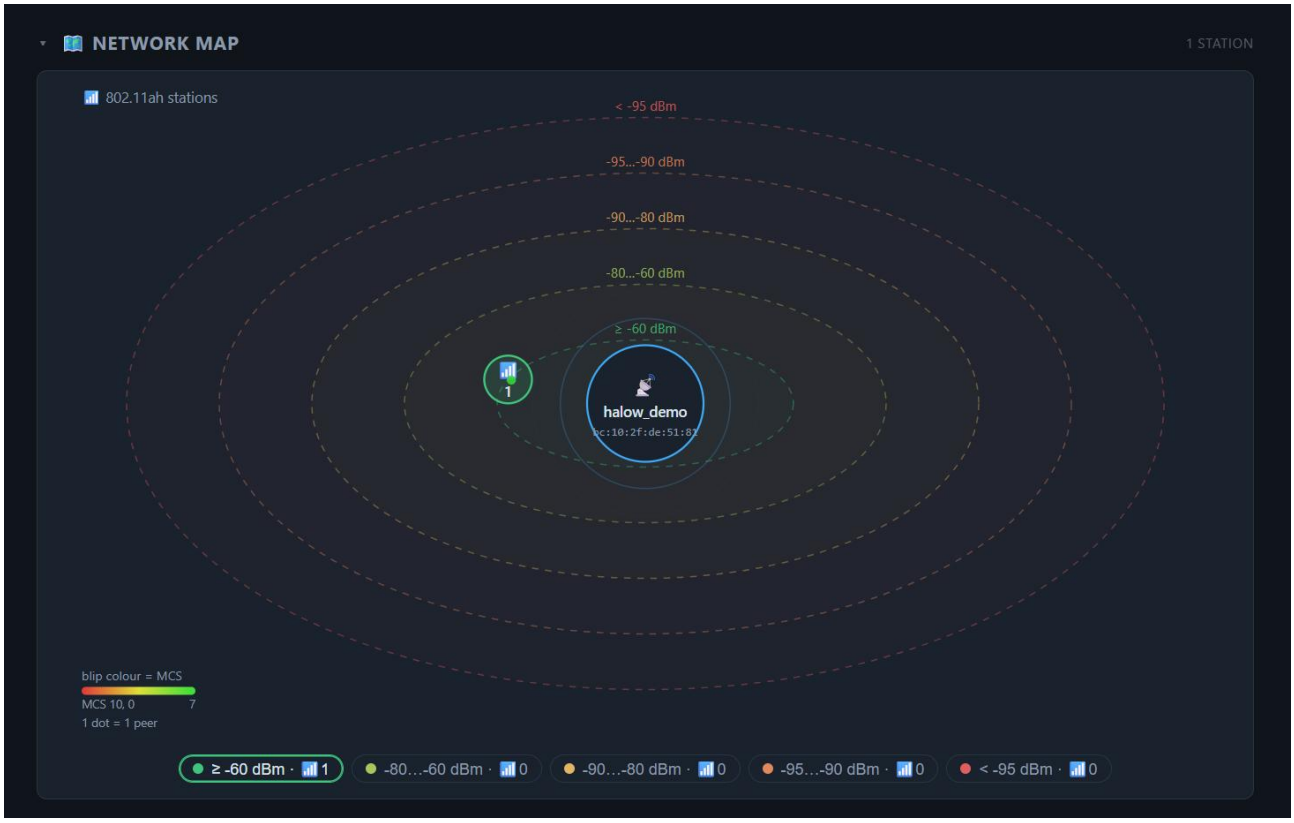



**Figure 5.4 AP network map with a station in an RSSI band**

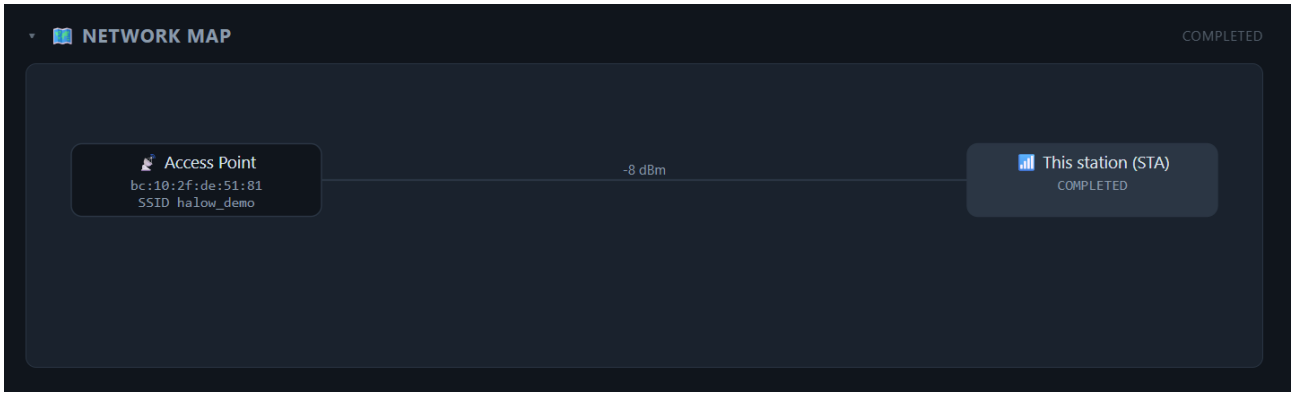



**Figure 5.5 STA network map showing the upstream AP**

Even when a device appears in the list, run **Ping** to check actual connectivity. Ping checks the path from the board to that device.

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## 5.3 Using the diagnostic tools

**Table 5.2 Diagnostic tools**

| Tool | Purpose |
|---|---|
| Ping | Check connectivity to a target address |
| Bridge clients | List the devices discovered behind the bridge |
| Traffic | Check the short-term throughput of the HaLow interface |
| CLI runner | Run supported CLI commands without an SSH session |
| Support bundle | Collect settings, logs and state for technical support |

The CLI runner supports read commands such as `show ...` and `help` by default. A few exceptions, `set chmgr`, `hlog` and `dump run`, are also available. `set chmgr` in particular changes behavior and can lead to an automatic channel switch, so read the confirmation text before running it.

General write commands are blocked by default. On a debug package the write restriction can be lifted for the current session only. For commands the CLI runner does not support, use the command-line environment the package provides, such as SSH.

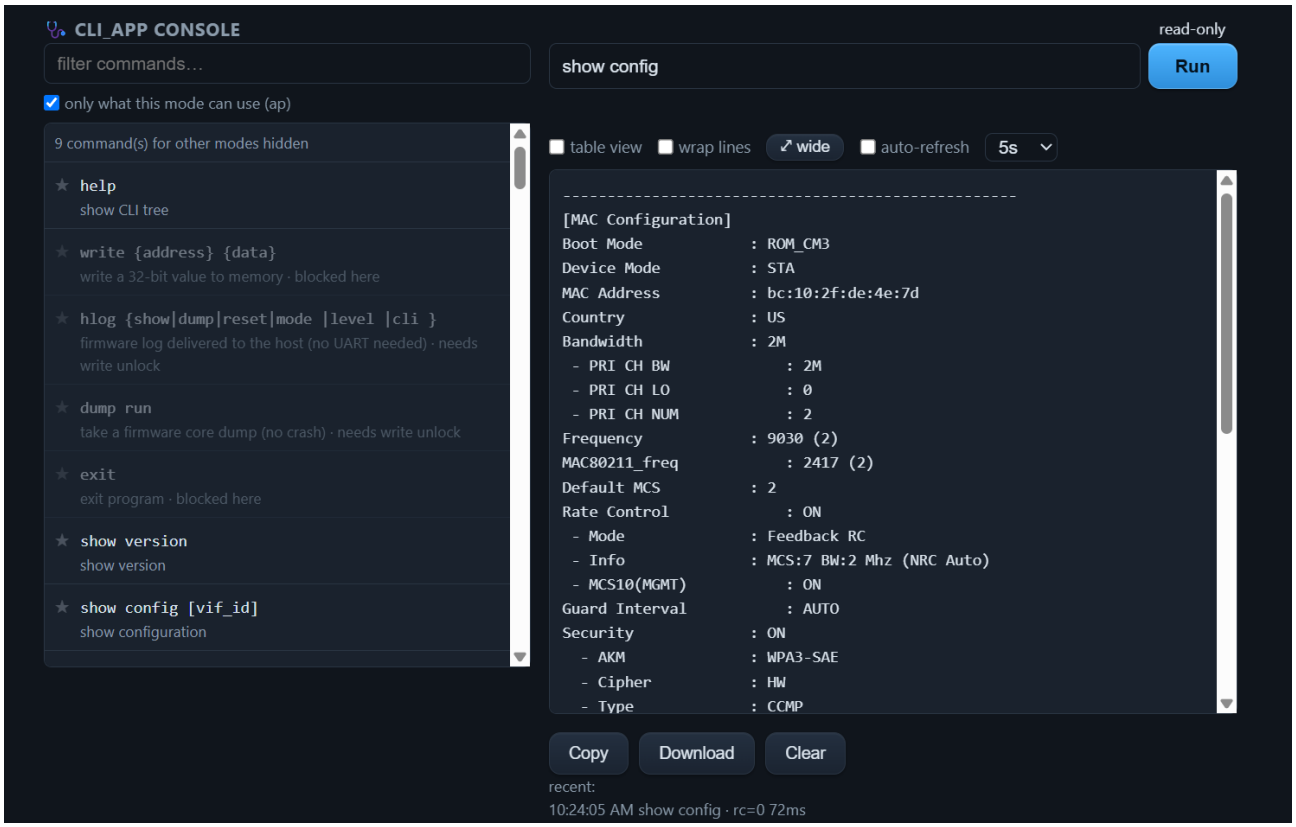



**Figure 5.6 CLI console with a command result**

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# 6 Backup and factory defaults

## 6.1 Exporting the settings

1. Select **Export** under **Settings**.
2. Keep the downloaded JSON file.

The export file contains the saved settings, the export time and the firmware version. It does not contain the administrator credentials.

## 6.2 Importing the settings

1. Back up the current settings with **Export** first.
2. Select the JSON file to import under **Import**.
3. Check the applied items and the skipped items in the result.
4. After the import, check the operating mode, the country, the channel and the network settings again.
5. Check the run state and traffic on the dashboard.

The administrator credentials and the wired management address are not imported, and unsupported items or items that fail validation are skipped. Do not assume the whole file is restored as it is.

Depending on the imported items, a wireless restart can occur. Unlike the Settings screen, the import does not check every combination of settings, so make sure there is no combination that cannot be used together, in particular a fixed channel with automatic selection.

## 6.3 Restoring factory defaults

1. Back up the settings you need with **Export**.
2. Run **Factory defaults** under **Settings**.
3. Check the result and enter the settings you need for operation again.
4. For an AP, select a channel, or `auto` where it is supported.
5. Check whether the wireless function is running and apply the state you want.

**The administrator credentials and the wired management address are kept** after restoring factory defaults. This operation cannot reset a forgotten password.

Restoring factory defaults restores the settings only; it does not start or stop the wireless function. The factory defaults contain no channel, so a channel must be set before an AP is started.

The settings the GUI manages are also written back to the related scripts and configuration files as defaults. The name of the backup file created on the board is shown in the result.

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# 7 Troubleshooting

**Table 7.1 Web GUI troubleshooting**

| Symptom | What to check and do |
|---|---|
| The GUI cannot be reached | Check the wired connection and the board management address, and connect to `https://<board address>:8443`. |
| A certificate warning is shown | This can be the board's self-signed certificate. Confirm the address you are connecting to. |
| `change the admin password first` is shown | Complete the administrator password change under System in Settings first. |
| A setting you want is not shown | Check whether it is supported in the current operating mode. |
| Saving the channel is rejected | Check that a fixed channel and `auto` were not set together, and that automatic selection is supported in this configuration. |
| Saved, but the wireless function does not start | `settings-only` means only the save completed. Start the wireless function from the dashboard. |
| The connection drops after a restart | Connect over the wired management address instead of the HaLow link. If you also changed the wired address, use the new one. |
| The AP does not start after restoring factory defaults | Set a channel, or `auto` where supported, then start it. |
| Only part of an imported file was applied | Check the skipped items in the import result and enter the settings you need on the screen. |
| A command is rejected in the CLI runner | Check whether it is a default read command or one of the supported exceptions. |

If the problem persists, collect the data with **Support bundle** and send it to technical support together with the time of the occurrence, the settings you changed and the error message on the screen.

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# 8 Revision history

| Version | Date | Description |
|---|---|---|
| Ver 1.0 | 9/28/2026 | Initial release |

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